

Assignment 3 THEORY OF COMPUTATION

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Ans-1) a) $\Sigma = \{a, b\}$
 $V = \{S, A\}$

Start Symbol : S
Productions (P):

$S \rightarrow aas / aaA$
 $A \rightarrow bA / \epsilon$

S generates an even number of a 's (at least 2)
and A generates any number of b 's.

(b) String : $aaaaabb$
Leftmost Derivation.

$S \rightarrow aas$
 $\rightarrow aa \ aaA$
 $\rightarrow aa \ aabA$
 $\rightarrow aa \ aab \ bA$
 $\rightarrow aa \ aa \ bb$

(c)

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      S
     / \
    aa  A
       |
       b
       |
       b
  
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Leaves: $aa \ aa \ bb \rightarrow "aaaaabb"$

(d) Regular Expression for L :

$$(aa) + b^*$$

Since this RE exists, L is a Regular Language and every regular language is also Context free.

Ans-2

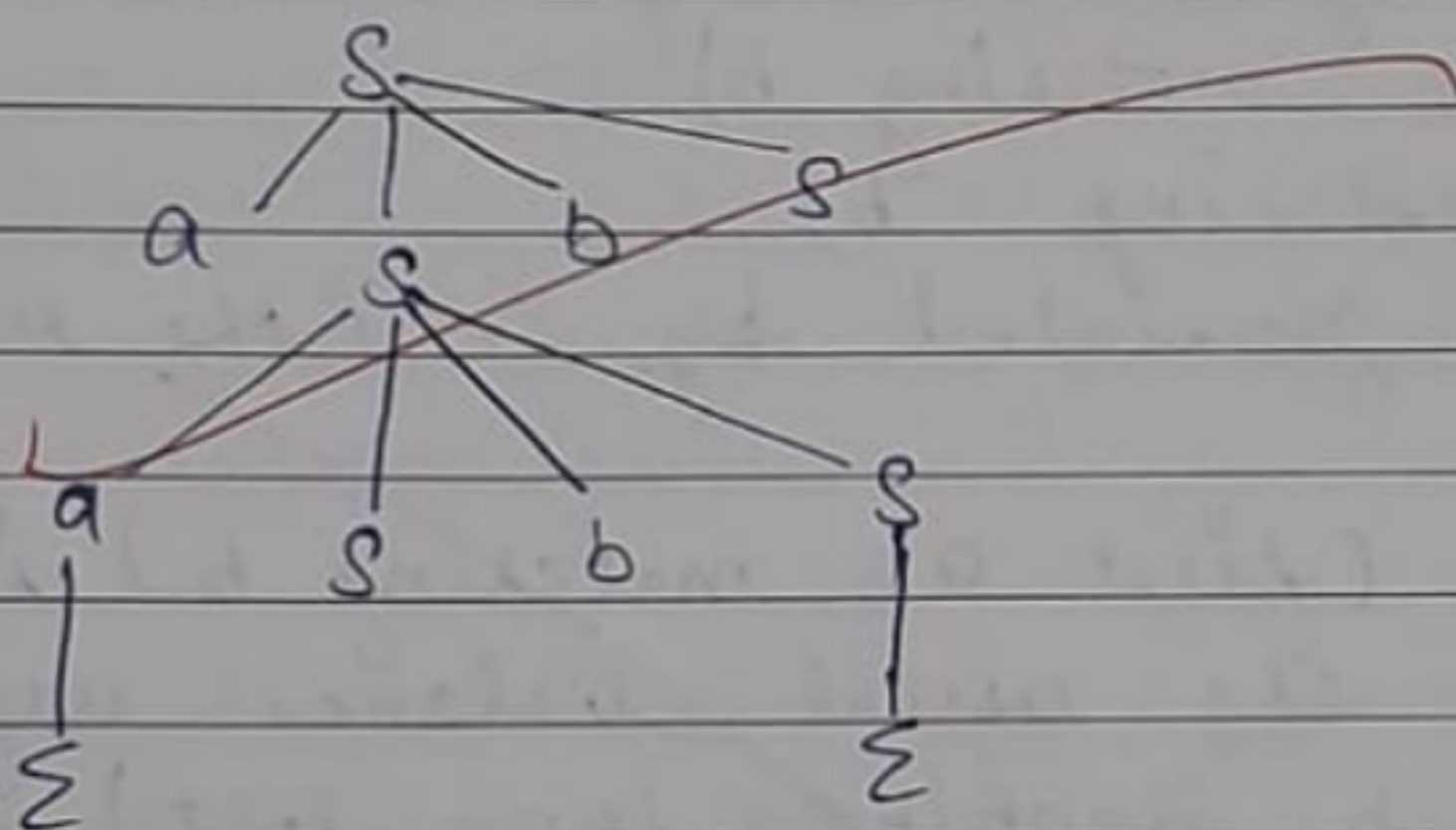
(a) Derive $aabb$

$$S \rightarrow asbs$$

$$\rightarrow a(asbs)bs$$

$$\rightarrow aa \in b \in b \in = aabb$$

Parse tree



leaves (left-to-right): $aa \ bb \rightarrow aabb$

(b) A grammar is ambiguous if there exists at least one string in the language that has two different leftmost or rightmost derivations.

claim: The grammar $S \rightarrow asbs | \epsilon$ is unambiguous

Proof: - Any non-empty string generated by this grammar must begin with a (because the only production introducing terminals start with a) and therefore must be produced by one application of $S \rightarrow aS, bS_2$

• In any derivation of a non-empty string N , the first a in N must match some b later in N . Let that matching b be at position K . The grammar forces that matching b to be the b produced in the same production $aS_1 bS_2$ that produced the first a .

- a (the first terminal)
- Substring generated by S_1 (between this a and its matching b)
- b (the matching b)
- Substring generated by S_2 (the remainder)

• The position K (first a 's matching b) is uniquely determined by the usual balance argument (Count a minus b scanning from left) - it is the smallest index where balance returns to the level before the first a . Hence the split into parts for S_1 and S_2 is unique.

• By induction on string length, the derivations (and therefore parse tree) of each of the two substrings are unique.

• Therefore the whole parse tree is unique.

So no string has two different parse trees
is unambiguous.

The given grammar itself is already non-ambiguous you can present an equivalent (and maybe clearer) unambiguous grammar.

$$S \rightarrow TS | \epsilon$$

$$T \rightarrow a S b$$

T generates one matched pair $a \dots b$ where the $\dots$ is a balanced substring (generated by S) and then $S \rightarrow TS$ allows concatenation of such matched blocks. This grammar is the same as the original $S \rightarrow a S b \mid \epsilon$ but written to emphasize the unique decomposition $S = (a S b)$ S - hence unambiguous.

Ans-3 Let $M = (Q, \Sigma, T, S, q_0, z_0, F)$ where.

$$\bullet Q = \{q_0, q_1, q_F\}$$

$$\bullet \Sigma = \{a, b\}$$

$$\bullet T = \{z_0, x\} - z_0 \text{ is initial stack symbol}$$

x is marker for each a .

$$\bullet q_0 \text{ is start state.}$$

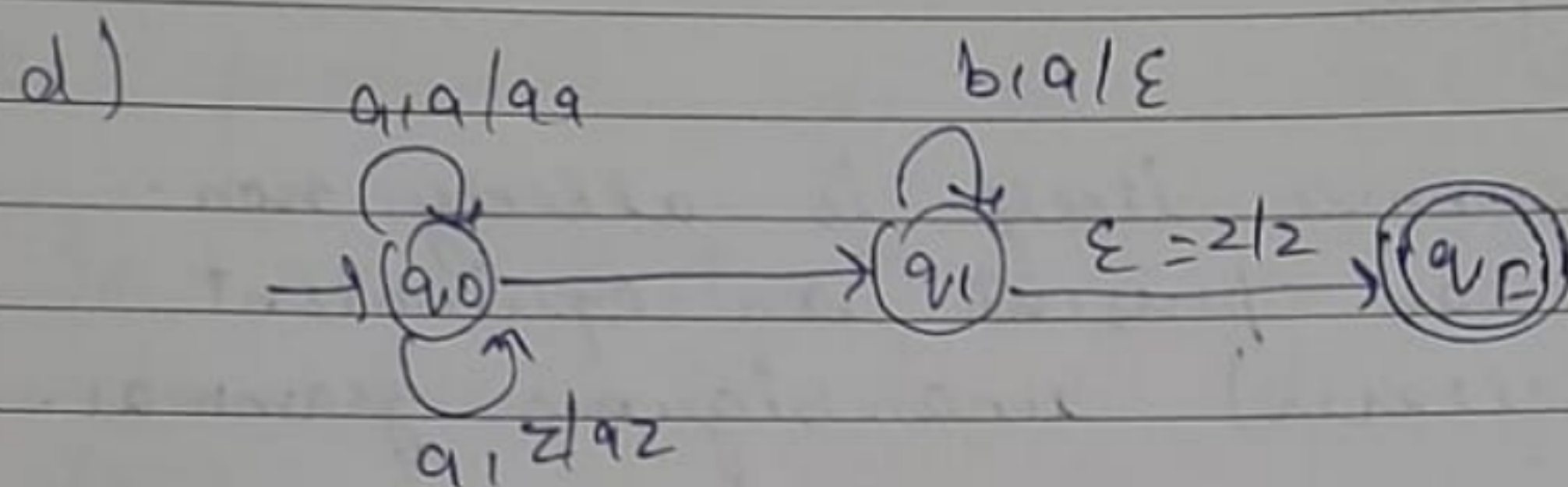
$$\bullet z_0 \text{ is initial stack symbol}$$

$$\bullet F = \{q_F\} \text{ (accept by final state)}$$

$$(b) \bullet S(q_0, a, z_0) = \{(q_0, x z_0)\}$$

$$\bullet S(q_0, \epsilon, z_0) = \{(q_F, z_0)\}$$

- $\delta(q_0, a, x) = \{ (q_0, xx) \}$
- $\delta(q_1, \epsilon, z_0) = \{ (q_1, z_0) \}$
- $\delta(q_0, b, x) = \{ (q_1, x) \}$



Ans-4

Step-1 - Find nullable non-terminals:

A is nullable (since $A \rightarrow \epsilon$).

B is nullable (since $B \rightarrow \epsilon$). S is not nullable.

Step-2 add productions obtained by omitting nullable occurrences.

From $S \rightarrow aA$: because A nullable, add $S \rightarrow a$

From $S \rightarrow bB$: because B nullable, add $S \rightarrow b$

From $A \rightarrow aA$: because A nullable, add $A \rightarrow a$

From $B \rightarrow bB$: because B nullable, add $B \rightarrow b$

Step-3 → remove original ϵ -productions: remove $A \rightarrow \epsilon$ and $B \rightarrow \epsilon$

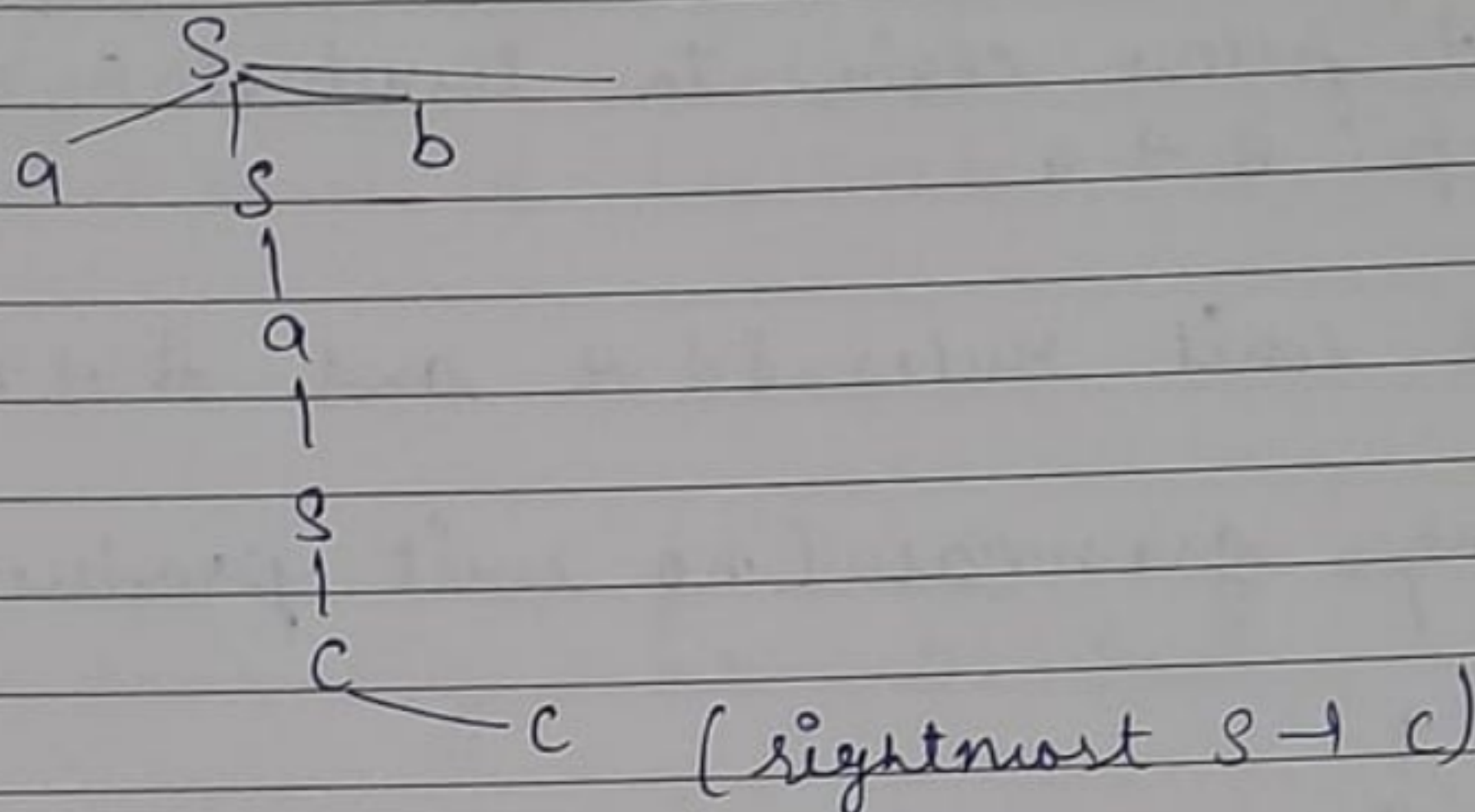
No unit productions (of the form $x \rightarrow y$) are present now.

b) CNF require productions of the form $x \rightarrow yz$.
(two - nonterminals) or $x \rightarrow a$ (Single terminal)
Also introduce helper non-terminals for terminals.

when they appear in longer right-hand sides.

$$x \rightarrow a \rightarrow x - a \rightarrow a$$

$$x \rightarrow b \rightarrow b$$



1) The right trees are structurally different (in A the left child of the root is 'a' then a subtree $asbs$; in B the root expands to $asbs$ immediately). Therefore the grammar is ambiguous (exists a string with two distinct parse trees).

2) A grammar is ambiguous if some string has two different parse trees. We exhibited two distinct parse trees for $acbc$. Hence the grammar is ambiguous.

Ques-8) Give the CFG: $G = \{ \{ S, A, B \}, \{ a \}, \{ S \rightarrow A, A \rightarrow B, B \rightarrow a \}, S \}$. Remove unit productions and rewrite the grammar.

Ans- Given CFG:
 $G = \{ \{ S, A, B \}, \{ a \}, P, S \}$ with
 $P: S \rightarrow A, A \rightarrow B, B \rightarrow a$.

Remove unit productions:
We have unit chains $S \rightarrow A \rightarrow B \rightarrow a$ and $A \rightarrow B \rightarrow a$.
Replace them by direct terminal productions:-

From S follow chain to terminal $a \Rightarrow$ add $S \rightarrow a$
From A follow chain to terminal $a \Rightarrow$ add $A \rightarrow a$
Keep $B \rightarrow a$

Remove unit rules $S \rightarrow A$ and $A \rightarrow B$

Resulting grammar (no unit productions)

$S \rightarrow a$

$A \rightarrow a$

$B \rightarrow a$

Note:- All non-terminals still generate a . The language is $\{a\}$

Ques-9 Give the CFG $G = (\{S, A, B\}, \{a, b, c\}, S)$. Remove useless productions and write the updated grammar.

Ans:- Given:

$G = (\{S, A, B\}, \{a, b, c\}, P, S)$ with
 $P: S \rightarrow A, A \rightarrow aB, B \rightarrow c$

Step-1) Find non terminals that generate terminals (useful for generating)

- $B \rightarrow c \Rightarrow B$ generates
- $A \rightarrow aB$ and B generates $\Rightarrow A$ generates
- $S \rightarrow A$ and A generates $\Rightarrow S$ generates.

So all S, A, B are generating.

Step-2) - Find reachable non-terminals from start S :

- S is start (reachable)
- From $S \rightarrow A \Rightarrow A$ reachable
- From $A \rightarrow aB \Rightarrow B$ reachable

So, all S, A, B are reachable.

$\Rightarrow$ There are no useless production (no unreachable or non-generating non-terminals).

* terminal grammar (after removing useless productions):

$S \rightarrow A$

$A \rightarrow aB$

$B \rightarrow C$

Language produced: strings of the form ac (specifically only " ac ")

Ques-10) Convert the given CFG to CNF. Consider the given grammar G_1 :

$S \rightarrow aAaAB$

$A \rightarrow aBB \mid \epsilon$

$B \rightarrow Aa \mid b$

Ans:- Step-1 - Remove ϵ -productions.
 $A \rightarrow \epsilon$ is nullable $\Rightarrow$ update others

$$\begin{aligned} S &\rightarrow a|aA|B \\ A &\rightarrow aBB \\ B &\rightarrow Aa|a|b \end{aligned}$$

Step-2) Remove unit productions:
 $S \rightarrow B$ $\rightarrow$ replace with B's R.H.S.

$$\begin{aligned} S &\rightarrow a|b|aA|Aa \\ A &\rightarrow aBB \\ B &\rightarrow Aa|a|b \end{aligned}$$

Step-3 - convert to CNF form.

Introduce $x \rightarrow a \rightarrow a$, $y \rightarrow BB$.

Replace terminals in long R.H.S and make binary:

$$\begin{aligned} S &\rightarrow a|b|x-aA|Ax-a \\ A &\rightarrow x-a y \\ y &\rightarrow BB \\ B &\rightarrow Ax-a|a|b \\ x-a &\rightarrow a \end{aligned}$$

Final CNF: All rules are either
 $A \rightarrow BC$ or $A \rightarrow a$.